

Emerging Technologies for the Circular Economy

Lecture 0: Organization

Prof. Dr. Benjamin Leiding
M.Sc. Anant Sujatanagarjuna
M.Sc. Chintan Patel

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- Updated versions of these slides will be available in our [Github repository](#).

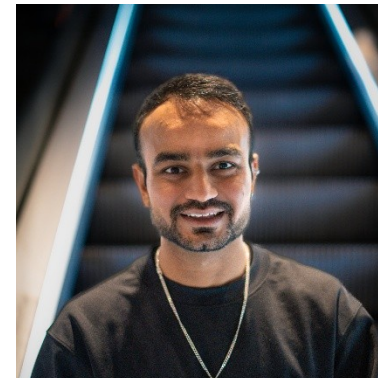
Team



Prof. Dr. Benjamin Leiding



M.Sc. Anant Sujatanagarjuna



M.Sc. Chintan Patel

ETCE Research Group

- **Emerging Technologies for the Circular Economy → ETCE**
- Research focus:
 - Intersection of IT and sustainability
 - Circular Economy
 - Self-organized, decentralized and distributed systems
 - Sustainable and resilient food production
- Other courses:
 - The Limits to Growth – Sustainability and the Circular Economy (WS – open for everyone)
 - Requirements Engineering (WS – M.Sc.)



ETCE Research Group

- ETCE Website – [Link](#)
 - Course material
 - Theses/project topics

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You want join us? Write us an email!

→ benjamin.leiding@tu-clausthal.de

Learning Outcome

- Basic understanding of the concept of the circular economy
- Basic understanding of emerging technologies at the intersection of digital technologies and sustainability
- Understanding and overview of the Internet of Things and related concepts
- Ability to design smart systems and applications in the context of connected sensor systems and other stakeholders.
- Experience in prototyping applications and systems

Lectures

- 27.04.2026 → Organization (L00)
- 04.05.2026 → Introduction (L01)
- 11.05.2026 → Circular Economy (L02)
- 18.05.2026 → Lifecycle Assessment – LCA I (L03)
- 01.06.2026 → Lifecycle Assessment – LCA 2 (L04)
- 08.06.2026 → Introduction to the Internet of Things (L05)
- 15.06.2026 → Internet of Things – Communication (L06) + Security and Privacy (L07)
- 22.06.2026 → Internet of Things – Data Processing and Big Data (L08)
- 29.06.2026 → IoT Applications I – Resilient and Sustainable Food Production (L09)
- 06.07.2026 → IoT Applications II – Biodiversity Monitoring (L10)
- 13.07.2026 → Digital Technologies and Greenwashing (L11)

- 20.07.2026 → Exam Q&A

Exercise Deadlines

- 11.05.2026 → Exercise 01 – Knowledge Test (MC)
- 18.05.2026 → Exercise 02 – Circular Economy (MC)
- 01.06.2026 → Exercise 03 – Lifecycle Assessment (LCA) – Part 1
- 08.06.2026 → Exercise 04 – Lifecycle Assessment (LCA) – Part 2
- 15.06.2026 → Exercise 05 – IoT Sensing and Gathering Data
- 22.06.2026 → Exercise 06 – IoT Data Processing
- 29.06.2026 → Exercise 07 – IoT Security
- 06.07.2026 → Exercise 08 – TBA
- 13.07.2026 → Exercise 09 – TBA
- 20.07.2026 → Exercise 10 – TBA

Course Organization

- Organization of the lecture:
 - Slides are available on Github ([Link](#))
 - Please report bugs!
 - Lectures and exercises as live stream (BBB – next slide)
 - Lecture recordings will be available on StudIP and on Github
 - Exercise time slots = Time for questions and eventual tutorials related to the exercises

Questions? Write us an email: etce-etce@tu-clausthal.de ← **We will only respond to emails written to this specific email address!**

Dates/Times/Locations

- Lecture:

- Monday **2:15 pm to 3:45 pm** (Berlin time) – **27.04.2026 to 13.07.2026**
- Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton ([Link](#))

- Exercise / Q&A:

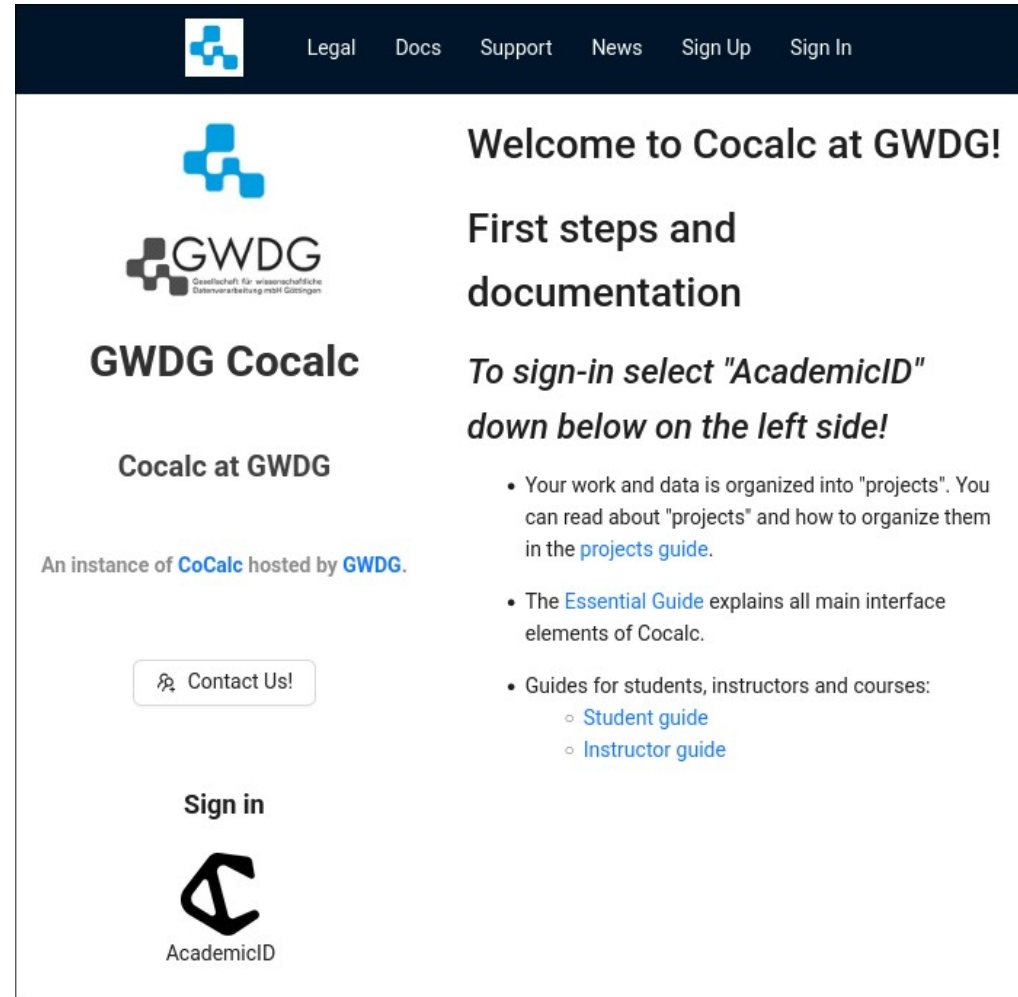
- Monday **4 pm to 4:45 pm** (Berlin time) – **27.04.2026 to 13.07.2026**
- Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton ([Link](#))

Exercises

- Individual work → no group submissions
- Multiple-Choice or coding exercises
- Multiple-Choice exercises are considered “self-evaluations” and are not submitted
- 7-14 days to submit (depending on the task)
- Submission deadline (coding exercises) is always Monday at 1:59pm (right before the next lecture)
- Submission of **each** coding exercise is **mandatory**
- **You pass by receiving a grade greater than 0.**
- You will receive feedback on your submission
- Exercise = learning feedback

Coding Exercise Submission and Grading

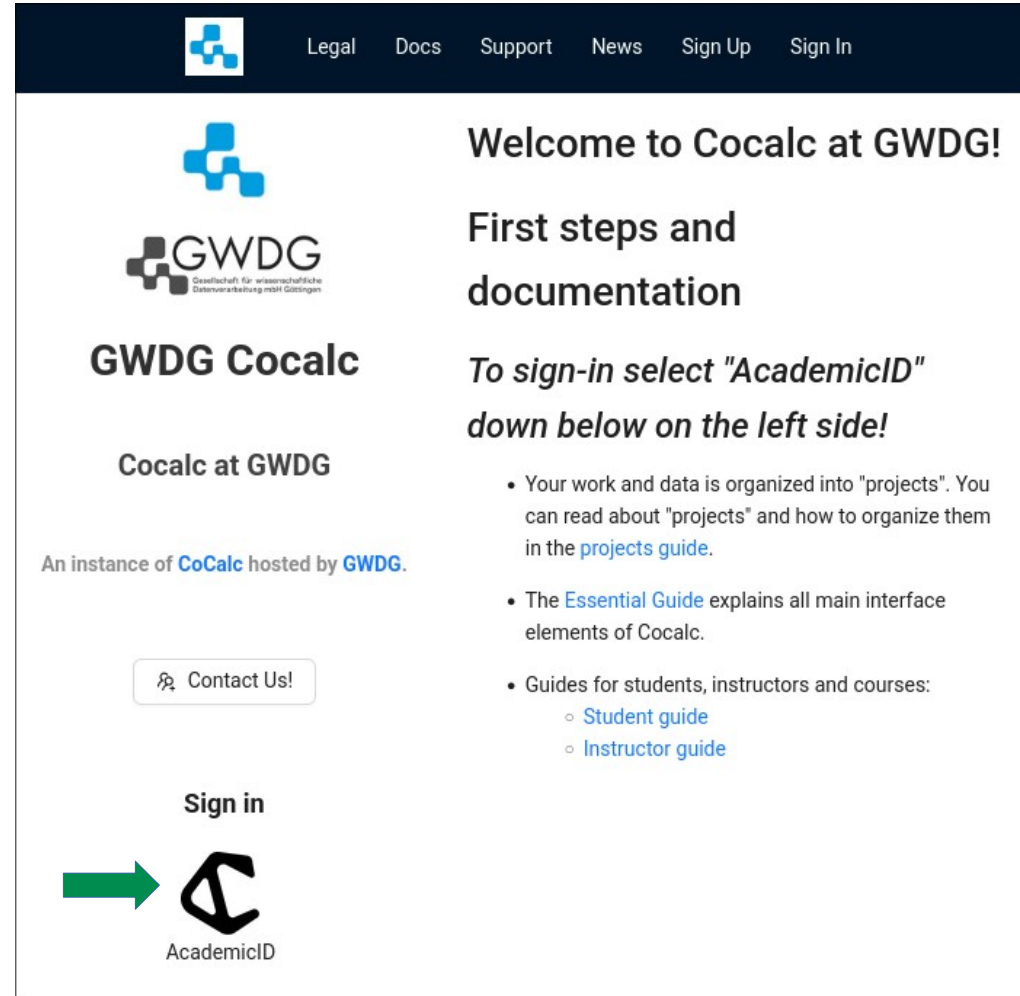
- Coding exercises are graded semi-automatically. Due to this it is highly important that you follow the required submission format. Otherwise the grading process will fail and you will receive 0 points.
- Coding exercises will be provided in the form of Jupyter Notebooks, and provided to you via Cocalc <https://cocalc.gwdg.de/>, with additional instructions provided via [GitHub](#).



The screenshot shows the homepage of Cocalc at GWDG. The header is dark blue with a logo and navigation links: Legal, Docs, Support, News, Sign Up, and Sign In. The main content area is white. On the left, there is a GWDG logo (Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen) and the text "GWDG Cocalc" and "Cocalc at GWDG". Below this, it says "An instance of CoCalc hosted by GWDG." and there is a "Contact Us!" button. At the bottom left is the "Sign in" button with the AcademicID logo. On the right, there is a welcome message "Welcome to Cocalc at GWDG!" followed by "First steps and documentation". Below this, it says "To sign-in select 'AcademicID' down below on the left side!". There are three bullet points: "Your work and data is organized into 'projects'. You can read about 'projects' and how to organize them in the [projects guide](#).", "The [Essential Guide](#) explains all main interface elements of Cocalc.", and "Guides for students, instructors and courses:" with sub-points for "Student guide" and "Instructor guide".

Coding Exercise Submission and Grading

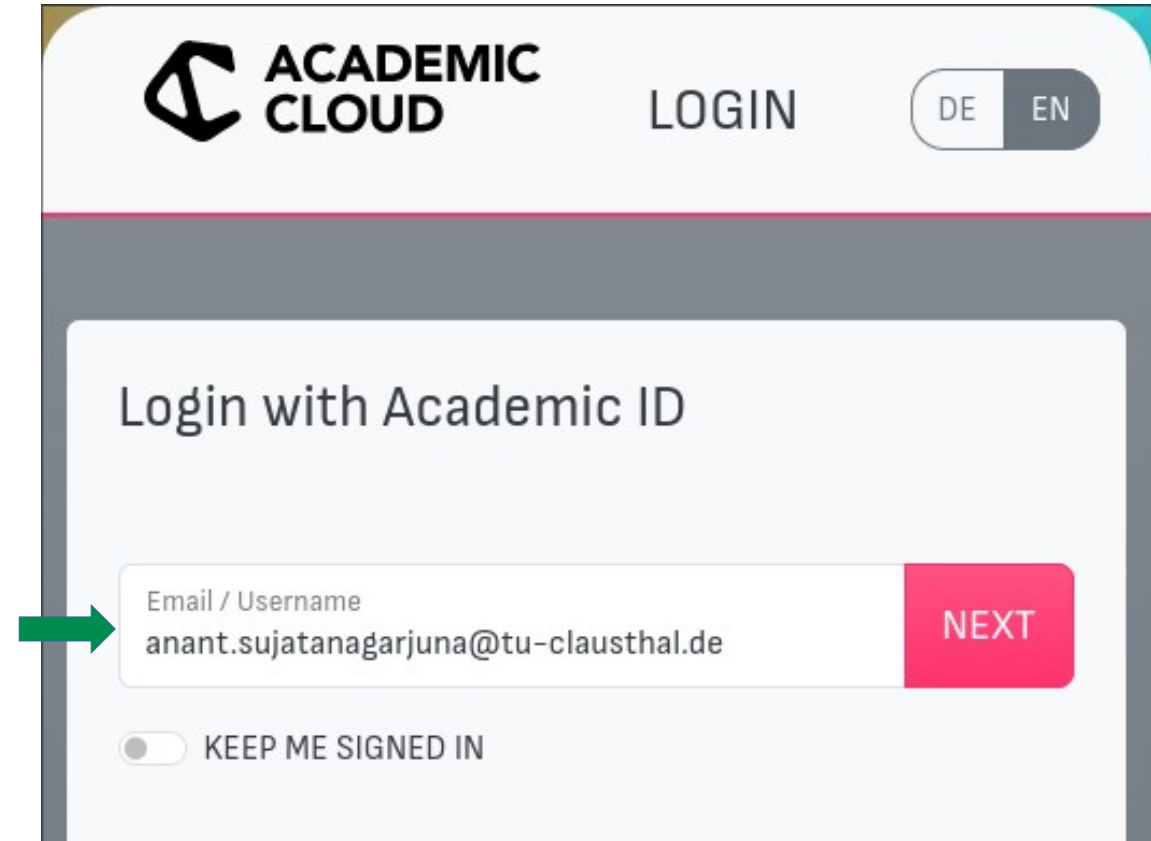
- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo



The screenshot shows the GWDG Cocalc website. At the top is a dark blue navigation bar with links: Legal, Docs, Support, News, Sign Up, and Sign In. Below the navigation bar, the GWDG logo is displayed, followed by the text "GWDG Cocalc" and "Cocalc at GWDG". A sub-header reads "An instance of CoCalc hosted by GWDG." Below this is a button labeled "Contact Us!". The main content area is split into two columns. The left column contains the "Sign in" text and the AcademicID logo, which is a stylized 'A' with a green arrow pointing to it from the left. The right column contains the text "Welcome to Cocalc at GWDG!", "First steps and documentation", and "To sign-in select 'AcademicID' down below on the left side!". Below this text is a bulleted list of links: "Your work and data is organized into 'projects'. You can read about 'projects' and how to organize them in the [projects guide](#).", "The [Essential Guide](#) explains all main interface elements of Cocalc.", and "Guides for students, instructors and courses:" with sub-links for "Student guide" and "Instructor guide".

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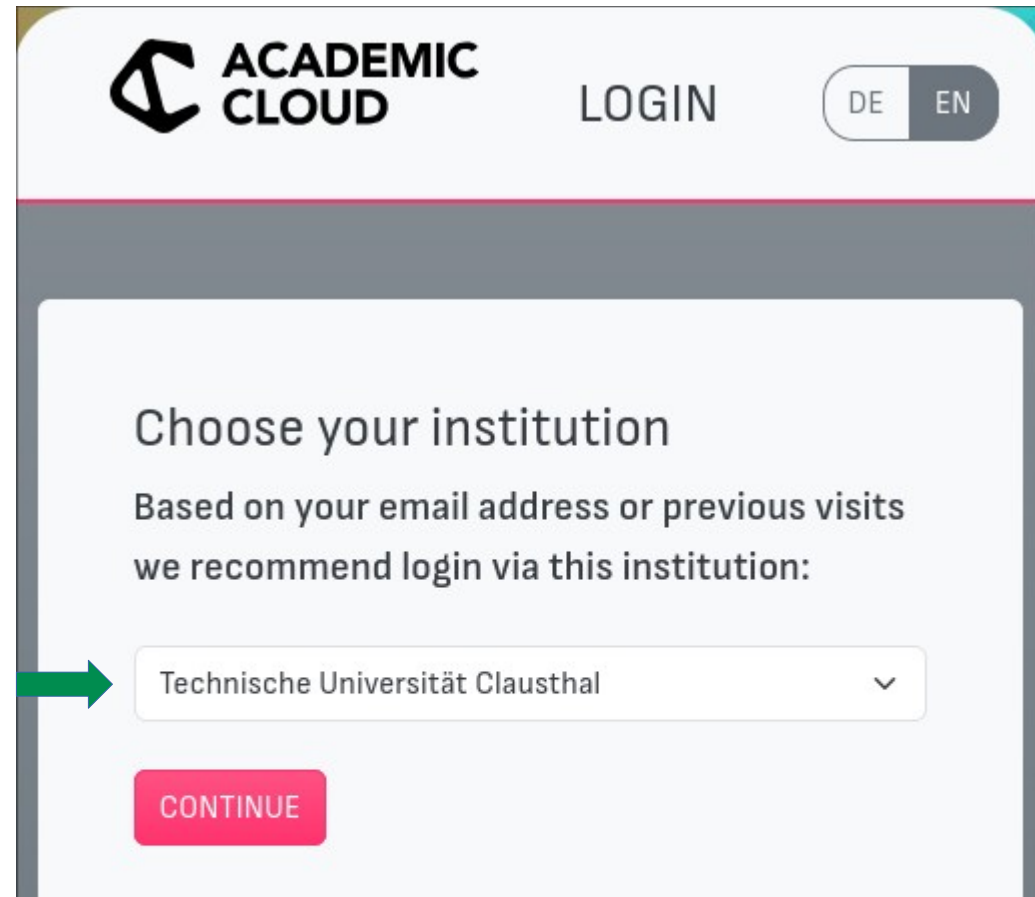
- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.



The image shows a web interface for 'ACADEMIC CLOUD'. At the top, there is a logo for 'ACADEMIC CLOUD' and a 'LOGIN' button. To the right of the login button is a language toggle switch with 'DE' and 'EN' options. Below the header, the main heading is 'Login with Academic ID'. There is a text input field labeled 'Email / Username' containing the email address 'anant.sujatanagarjuna@tu-clausthal.de'. A green arrow points to this input field. To the right of the input field is a red button labeled 'NEXT'. Below the input field, there is a toggle switch for 'KEEP ME SIGNED IN' which is currently turned off.

Coding Exercise Submission and Grading

- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.
 3. Choose “Technische Universität Clausthal” from the drop-down box.



ACADEMIC CLOUD LOGIN DE EN

Choose your institution

Based on your email address or previous visits we recommend login via this institution:

Technische Universität Clausthal

CONTINUE

Coding Exercise Submission and Grading

- Sign-in using your university credentials.
 1. Click on the “Academic ID” Logo
 2. Enter your university email address.
 3. Choose “Technische Universität Clausthal” from the drop-down box.
 4. Login with your usual TU-Clausthal credentials.



TU Clausthal
Anmelden bei Academic Cloud



Academic Cloud

Benutzername

Passwort

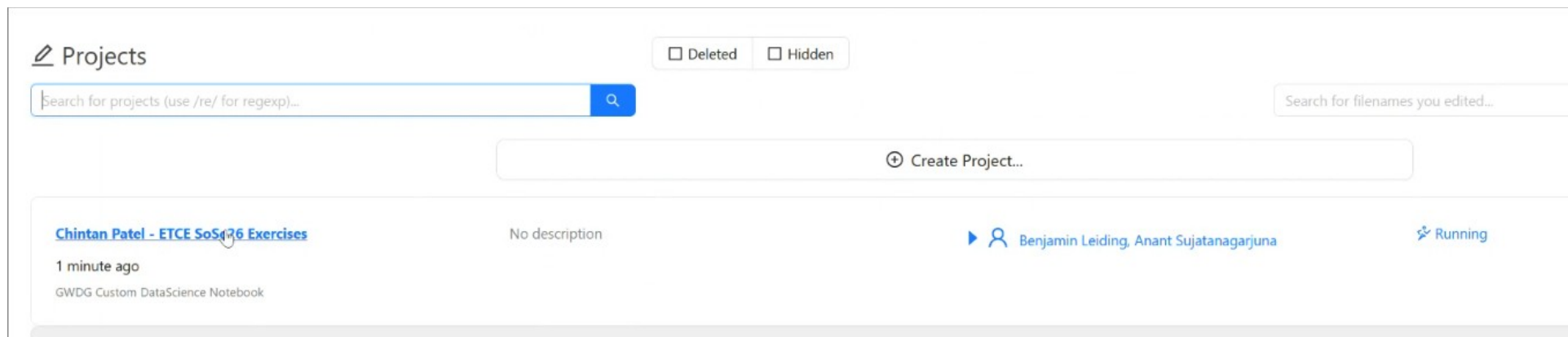
☐ Anmeldung nicht speichern

☐ Die zu übermittelnden Informationen anzeigen, damit ich die Weitergabe gegebenenfalls ablehnen kann.

Anmelden

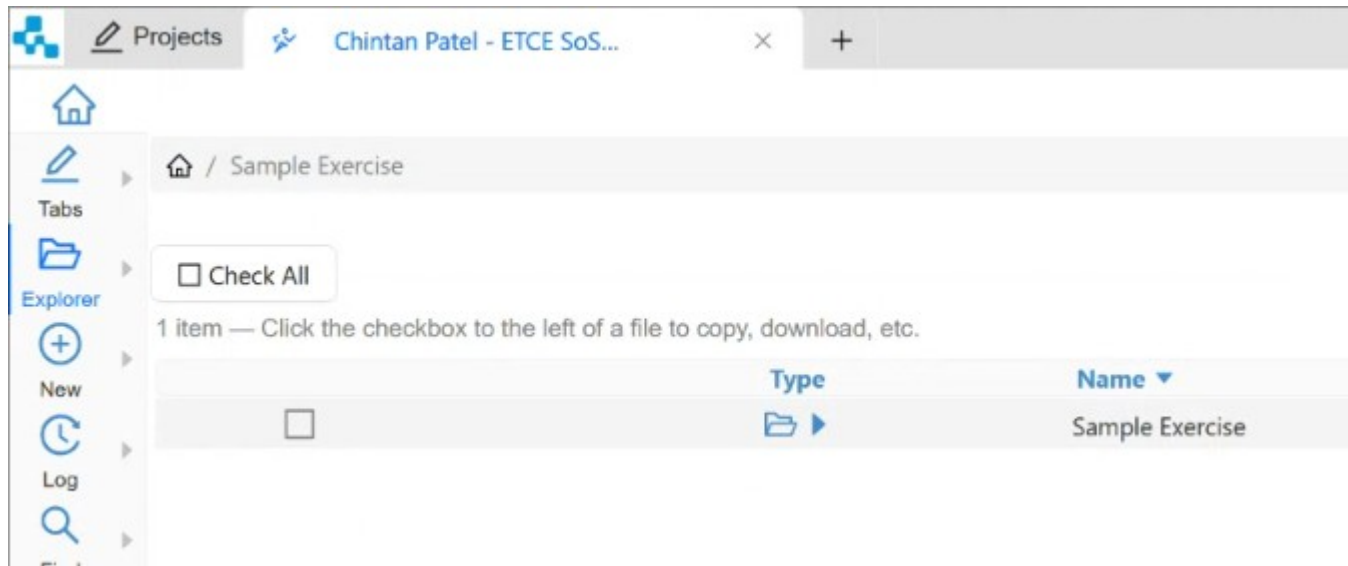
Coding Exercise Submission and Grading

- Submitting Assignments
 1. Click on “ETCE SoSe26 Exercises”



Coding Exercise Submission and Grading

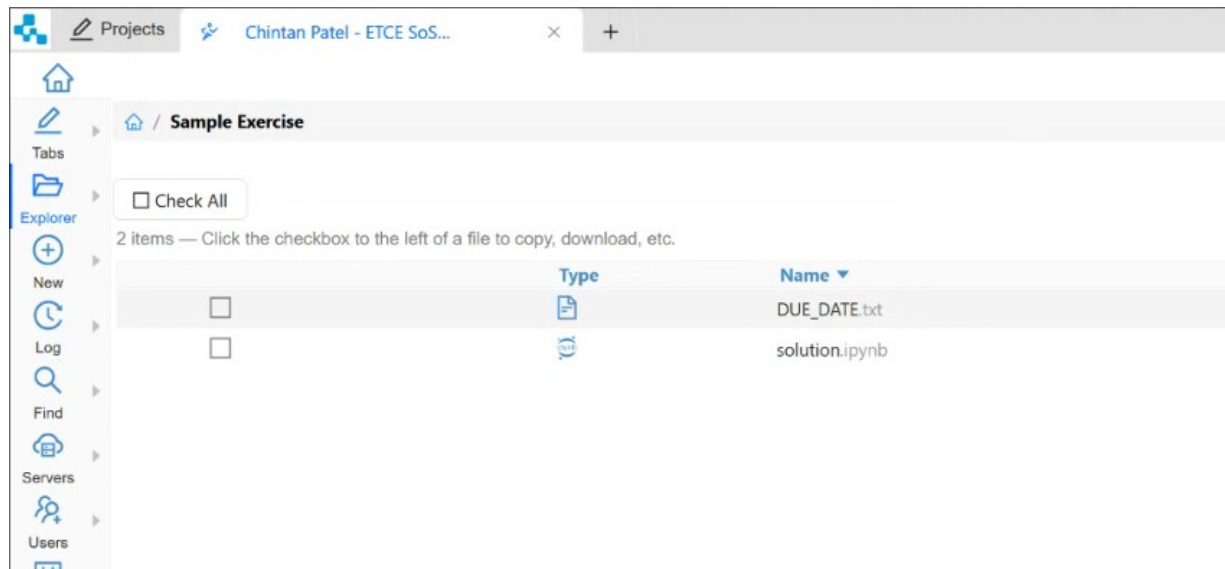
- Submitting Assignments
 1. Click on “ETCE SoSe26 Exercises”
 2. Open the folder of the exercise you want to work on.



Coding Exercise Submission and Grading

■ Submitting Assignments

1. Click on “ETCE SoSe26 Exercises”
2. Open the folder of the exercise you want to work on.
3. Edit the file in the exercise folder as indicated in the exercise handout PDFs.



Coding Exercise Submission and Grading

■ Submitting Assignments

1. Click on “ETCE SoSe26 Exercises”
2. Open the folder of the exercise you want to work on.
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The screenshot shows a Jupyter Notebook interface with a sidebar on the left containing icons for Log, Find, Servers, Users, Upgrades, Processes, and Settings. The main area is titled "Sample Programming Exercise" and contains two code cells. The first cell, labeled "In [24]:", contains a Python function definition for `get_the_date()` that returns the current date. The second cell, labeled "In [23]:", contains a Python function definition for `evaluate()` that checks if the output of `get_the_date()` is a `datetime.date` object and returns a grade of 100 if it is, or 0 otherwise. The output of the second cell is displayed as "Out[23]: Your grade is: 100%".

```
Log
Find
Servers
Users
Upgrades
Processes
Settings

Sample Programming Exercise
Edit the code in the following code blocks to complete your submission for this programming assignment

In [24]:
import datetime
def get_the_date() -> datetime.date:
    """
    This function should return today's date.
    """
    pass

Once you are done writing your solution, you can evaluate your solution in the code cells below:

In [23]:
def evaluate():
    """
    Use this function to evaluate whether your code works, and to see what grade your code will receive.
    """
    if type(get_the_date()) == datetime.date:
        return 100
    else:
        return 0

    print("Your grade is: "+str(evaluate()))+"")

Out[23]: Your grade is: 100%
```

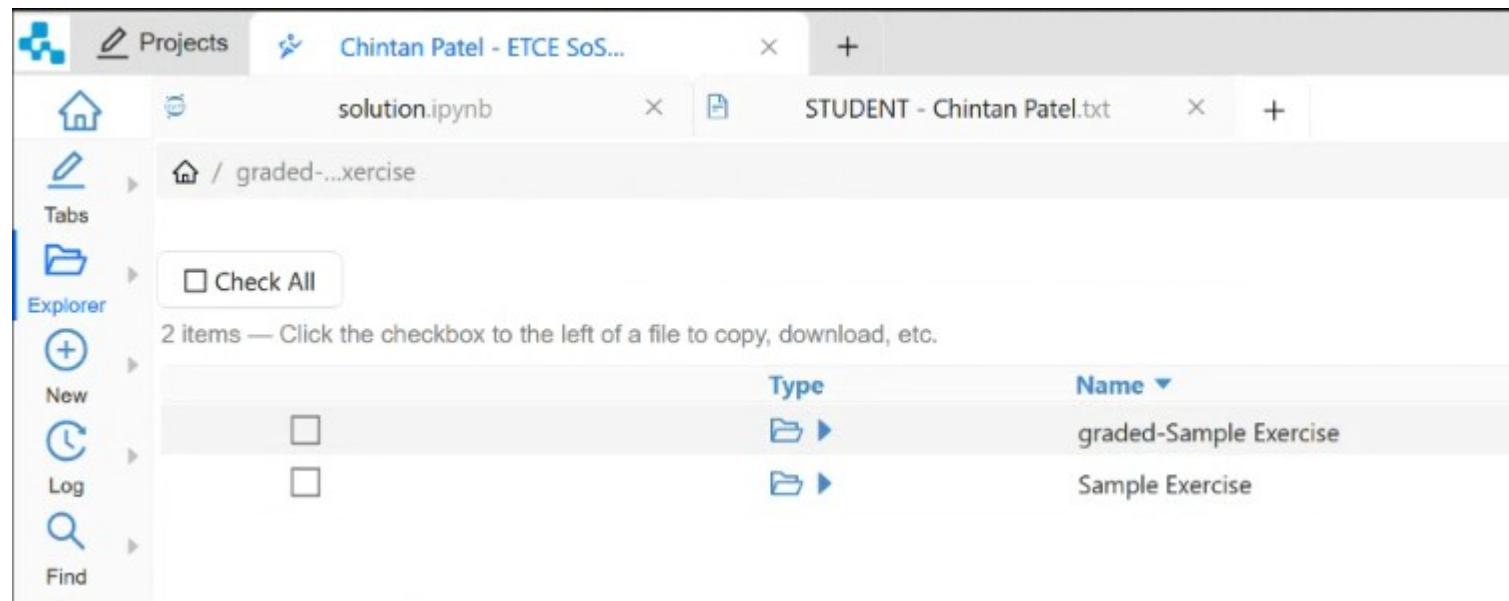
Coding Exercise Submission and Grading

- Exercise Grades
 - We will collect your exercises after the deadline has elapsed.
 - Please ensure to save all changes to the exercise folder BEFORE the deadline.
 - DO NOT submit your assignments via email/other means.

Coding Exercise Submission and Grading

■ Exercise Grades

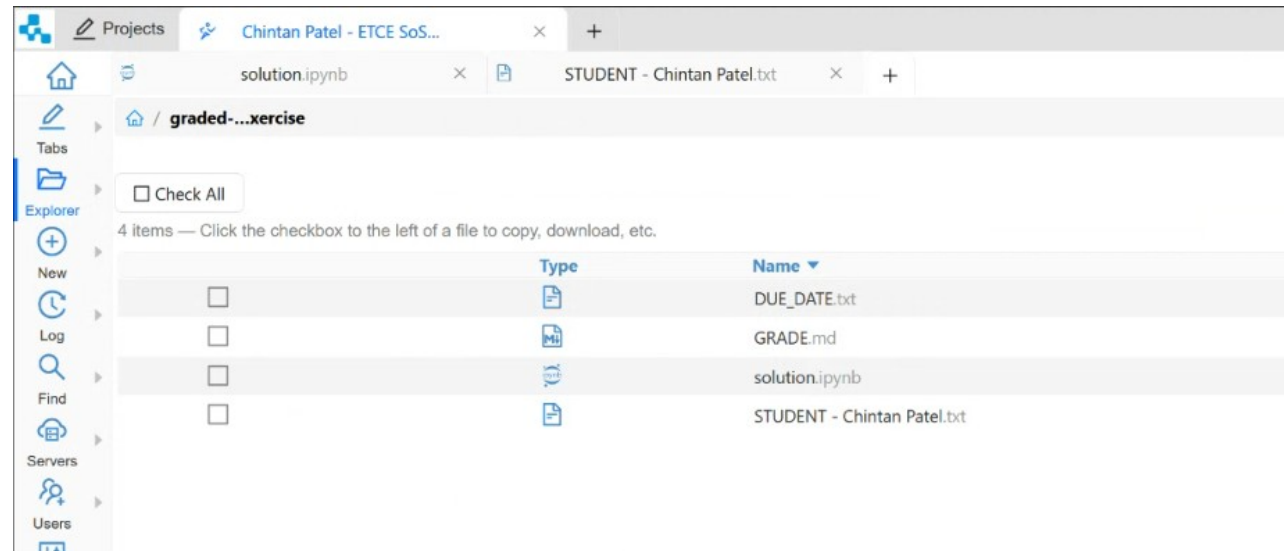
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- Exercises, once graded will be released back to you.



Coding Exercise Submission and Grading

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- Exercises, once graded will be released back to you.
- Your grade is provided inside the “GRADE.md” file.



Coding Exercise Submission and Grading

- Exercise Grades
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 - Please ensure to save all changes to the exercise folder BEFORE the deadline.
 - DO NOT submit your assignments via email/other means.
 - Exercises, once graded will be released back to you.
 - Your grade is provided inside the “GRADE.md” file.
 - Receiving a grade of 0 = FAIL.

Multiple-Choice Exercises

Every student enrolled in this course is advised to take the knowledge quiz in first two weeks of the course.

- **Goal of the test:**
 - To check the knowledge level of the student that is relevant to this course of study.
- **Preparation:**
 - A review of basic concepts of Cryptography and Circular Economy is recommended for Week 1 and Week 2 respectively
 - Knowledge quiz for Week 1 only tests your existing knowledge.
- **THE TEST IS JUST FOR YOU → WE CANNOT CHECK THE TEST RESULTS**

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Link for all multiple-choice exercises → etce.etce-lab.de

Multiple-Choice

E01: Prior Knowledge

Introduction

This question is the exercise for "*L01 - Introduction*" of the "Emerging Technologies for the Circular Economy" course taught at the TU Clausthal. All the course material is hosted on our [GitHub](#) page. This multiple-choice quiz tests your prior knowledge of the topics that we will discuss during the course.

You must answer at least half **the questions** correctly to pass the exercise.

If you have any questions, you can contact us via e-mail: etce-etce@clausthal.de

Question 01: Block ciphers are used in symmetric cryptography to encrypt and decrypt data. They operate on blocks of a fixed length (usually 64 or 128 bits). To encrypt some data, you could split it into block sized chunks and simply encrypt them separately. What which of the following properties are true?

Choose one or more options then select Submit.

- ☐ This is called ECB mode
- ☐ This is called CBC mode
- ☐ Patterns in the data will show up, which can impact security
- ☐ There is no integrity protection
- ☐ It will be easy to calculate the key

Submit

Multiple-Choice Exercises

Question 01: Block ciphers are used in symmetric cryptography to encrypt and decrypt data. They operate on blocks of a fixed length (usually 64 or 128 bits). To encrypt some data, you could split it into block sized chunks and simply encrypt them separately. What which of the following properties are true?

Choose one or more options then select Submit.

This is called ECB mode

This is called CBC mode

Patterns in the data will show up, which can impact security

✓ There is no integrity protection

It will be easy to calculate the key

Show correct answer



Examination

- Prerequisite for admission to the final exam (all criteria have to be fulfilled):
 - Submit all exercises (except for the MC exercises).
- Final exam:
 - Written exam (120min) via Moodle
 - Date → Most likely **03.08.2026 from 2 pm - 4 pm**

Self-Study Star

Self-Study Star → 

- Slides with the self-study star indicate optional/additional study material that is **not** mandatory but could be helpful or interesting

Literature

- This course is not based on a single book and you **do not** need to buy a book to pass the exam.
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *The Limits to Growth* (1972).
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *Limits To Growth: The 30-Year Update* (2004).
- Baccini et al. *Metabolism of the Anthroposphere: Analysis, Evaluation, Design* (2012).
- Walter R. Stahel. *The Circular Economy: A User's Guide* (2019).
- W. Brian Arthur. *The Nature of Technology: What It Is and How it Evolves* (2011).
- David Wallace-Wells. *The Uninhabitable Earth, Annotated Edition* (2017).
- (German) Stefan Rahmstorf, Hans Joachim Schellnhuber. *Der Klimawandel* (2019).

Literature

- M.A. Khan, M.T. Quasim, F. Algarni, A. Alharthi. *Decentralised Internet of Things* (2020).
- Dimitrios Serpanos und Marilyn Claire Wolf. *Internet-of-Things (IoT) Systems Architectures, Algorithms, Methodologies* (2018).
- Perry Lea. *Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security* (2018).
- Dan Boneh, Amit Sahai und Brent Waters. *Functional Encryption: Definitions and Challenges* (2010).

Questions?